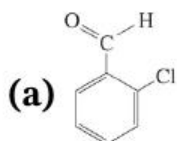


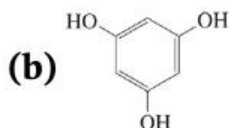
Solutions to Problems

36. (a) 3-Chlorobenzenecarboxylic acid, 3- or *m*-chlorobenzoic acid
(b) 1-Methoxy-4-nitrobenzene, 4- or *p*-nitroanisole
(c) 2-Hydroxybenzenecarbaldehyde, 2- or *o*-hydroxybenzaldehyde
(d) 3-Aminobenzenecarboxylic acid, 3- or *m*-aminobenzoic acid
(e) 4-Ethyl-2-methylbenzenamine, 4-ethyl-2-methylaniline
(f) 1-Bromo-2,4-dimethylbenzene
(g) 4-Bromo-3,5-dimethoxybenzenol, 4-bromo-3,5-dimethoxyphenol
(h) 2-Phenylethanol
(i) 1-Bromo-8-chloronaphthalene
(j) 1-(Phenanthren-3-yl)ethanone, 3-ethanoylphenanthrene, 3-acetylphenanthrene
37. (a) 1,2,4,5-Tetramethylbenzene
(b) 4-Hexyl-1,3-benzenediol
(c) 2-Methoxy-4-(2-propenyl)benzenol or 2-methoxy-4-(2-propenyl)phenol

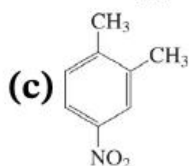
38.



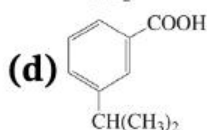
Name is acceptable. (Original IUPAC: 2-chlorobenzencarbaldehyde)



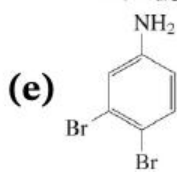
Name is numbered incorrectly; call it 1,3,5-benzenetriol.



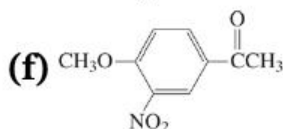
Name is incorrect. Never mix *o*, *m*, *p* with numbers; call it 1,2-dimethyl-4-nitrobenzene.



Name is acceptable. [Original IUPAC: 3-(1-methylethyl)-benzenecarboxylic acid]

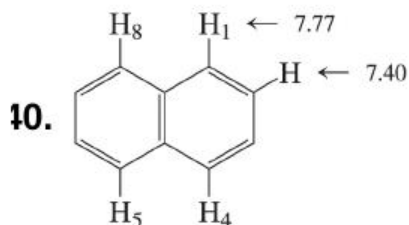


Wrong numbers; 3,4-dibromoaniline.



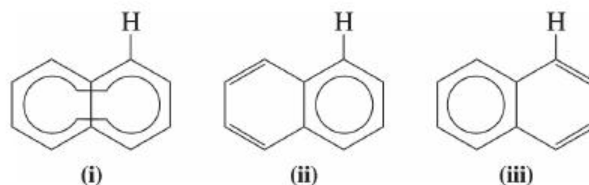
Use **only numbers**: 4-methoxy-3-nitroacetophenone or 1-(4-methoxy-3-nitrophenyl)ethanone.

39. Benzene would be higher in energy by about 30 kcal mol⁻¹, 30 kcal mol⁻¹, so ΔH_{comb} would be -819 kcal mol⁻¹.
-819 kcal mol⁻¹.



The hydrogens at carbons 1, 4, 5, and 8 are deshielded because they are closer to the **other** benzene ring in the molecule. They feel the deshielding effects of π -electron π -electron ring currents three ways: the

ring current around the whole molecule (i), the ring current of their own benzene ring (ii), and the ring current of the **adjacent** benzene ring (iii):



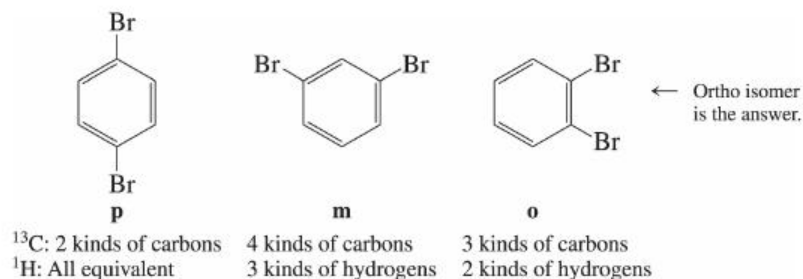
The hydrogens at carbons 2, 3, 6, and 7 are too far away to feel a significant amount of the ring current of the other benzene ring.

11. Yes. Cyclooctatetraene is described as lacking any special stabilization such as aromaticity. As a result, hydrogenation of its four double bonds should give off four times the energy that hydrogenating one double bond releases. The data indicate that this is approximately the case.

12. Rule: Aromaticity requires [a] $(4n+2) \pi$ electrons $(4n+2) \pi$ electrons contained in [b] a complete, unbroken circle of p orbitals. *p orbitals.*

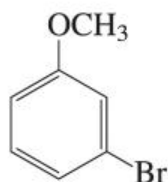
(a) No. 3 π electrons. *π electrons.* **(b)** Yes. Benzene is intact; extra double bond is an irrelevant substituent, not being part of the circle. **(c)** No. The saturated carbon is sp^3 , *sp^3* , breaking the circle of p orbitals; *p orbitals*; without a circle of p orbitals, *p orbitals*, the number of π electrons *π electrons* is irrelevant. **(d)** Yes. 10 π electrons; *π electrons*; the sp^3 *sp^3* carbon here is bridging and does not interrupt the p orbital *p orbital* circle. **(e)** No. 12 π electrons; *π electrons*; wrong number. **(f)** No. 9 π bonds = 18 π electrons, *9 π bonds = 18 π electrons*, which would be fine, except that the 2–2– charge adds 2 more for a total of 20 electrons; wrong number. **(g)** No. Saturated ring fusion carbons interrupt circle.

13. (a) UV spectrum supports the presence of benzene ring. ^{13}C NMR spectrum: three peaks, so the molecule must have symmetry. ^1H NMR spectrum: two sets of signals in equal intensity. Look at the three possible dibromobenzenes. The answer is clear:

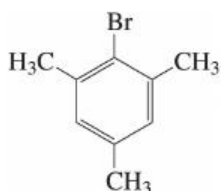


The IR (textbook page 709: single band at 745 cm^{-1} – 1745 cm^{-1}) agrees with the conclusion.

- (b) ^1H NMR spectrum: four benzene H's (one quite different from the other three), CH_3O —($\delta=3.7$). IR: meta disubstituted benzene. The answer is

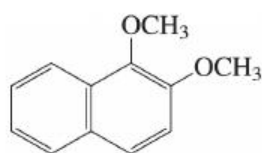


- (c) ^1H NMR spectrum: two benzene H's, three CH_3 's (CH_3 's (two equivalent; note just **two** methyl carbons in ^{13}C NMR spectrum, so the molecule has some symmetry to it. The answer (use trial and error) is

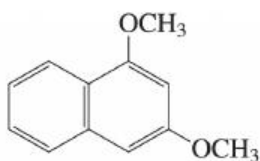


14. The top spectrum corresponds to the benzene derivative. Its general appearance, a strong molecular ion and few peaks of lower mass, reflects a compound that does not undergo ready fragmentation, a characteristic of aromatic compounds. The only significant fragmentation is loss of a single hydrogen atom to give the resonance-stabilized carbocation  (a system that will be discussed at length in [Chapter 22](#)). In contrast, the bottom mass spectrum shows extensive fragmentation, as one frequently sees with alkynes ([Section 13-3](#)). Note, for example, the strong peak at $m/z = 39$ for the resonance-stabilized cation $\text{HC}\equiv\text{CCH}_2^+$.

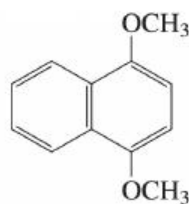
15. (a) Yes it is. Compare the answer to [Problem 43 \(a\)](#) and you'll see that each disubstitution pattern gives a unique number of $^{13}\text{C}^{13}\text{C}$ peaks for the ring carbons. Adding the peak for the two equivalent methoxy carbons in each molecule, we expect the para isomer to have three $^{13}\text{C}^{13}\text{C}$ peaks, the meta five, and the ortho four.
- (b) The ten possible isomers and the number of $^{13}\text{C}^{13}\text{C}$ peaks (ring carbons + methoxy carbons, which are no longer necessarily equivalent) for each are given below.



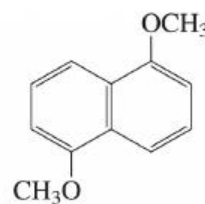
10 ring + 2 CH_3O



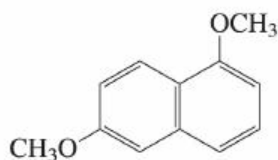
10 ring + 2 CH_3O



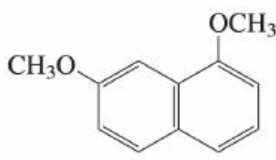
5 ring + 1 CH_3O



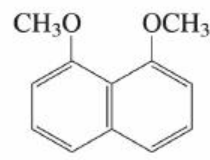
5 ring + 1 CH_3O



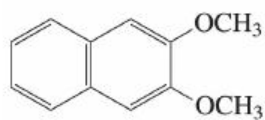
10 ring + 2 CH_3O



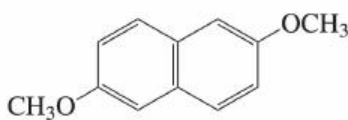
10 ring + 2 CH_3O



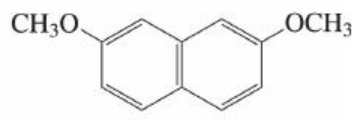
6 ring + 1 CH_3O



5 ring + 1 CH_3O

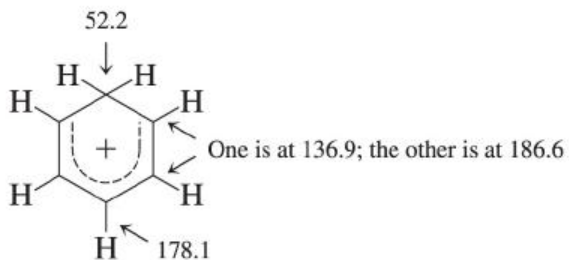


5 ring + 1 CH_3O

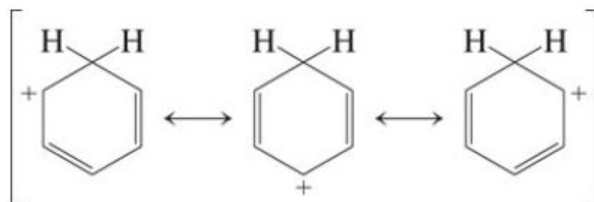


6 ring + 1 CH_3O

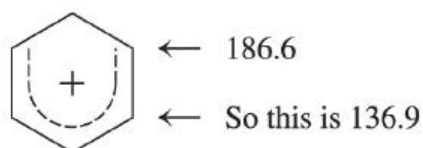
16.



Assign the signals at 136.9 and 186.6 ppm more precisely by looking at the resonance forms:

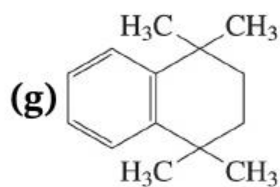
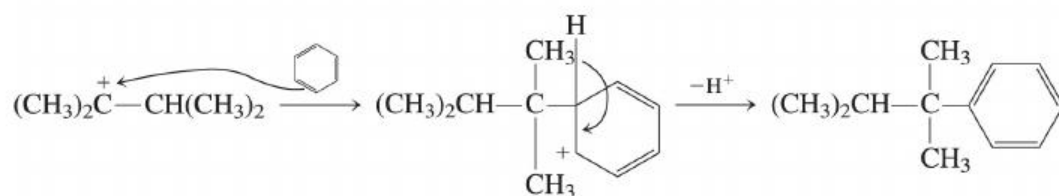
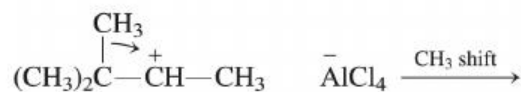
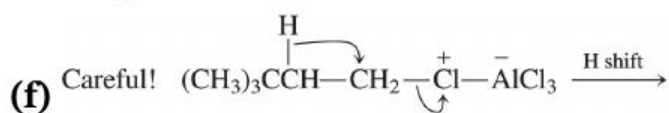
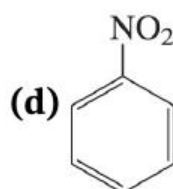
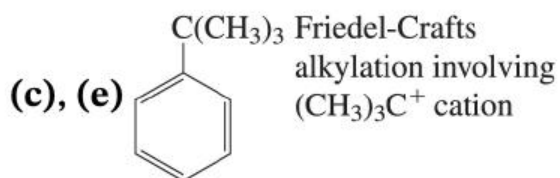
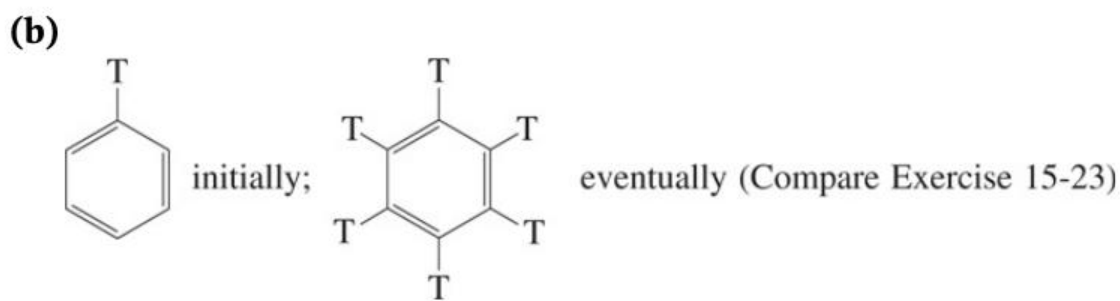
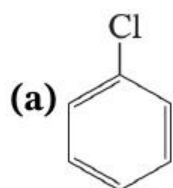


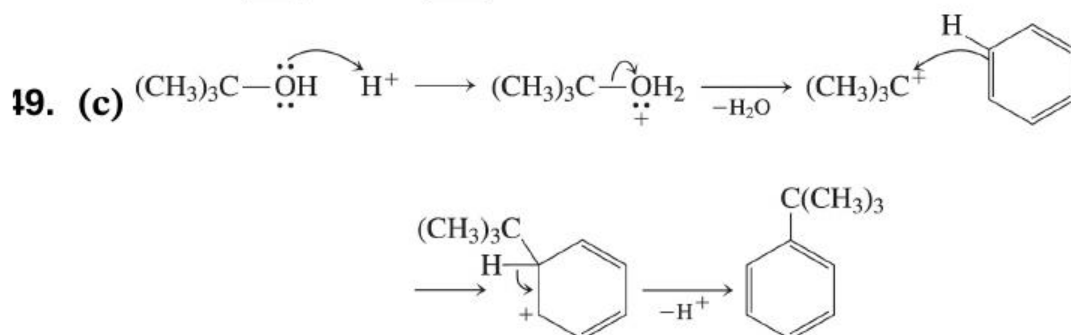
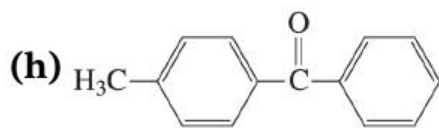
In the resonance forms, positive charges are located at only three of the carbons: They should be the most deshielded. That explains the $\delta = 178.1$ chemical shift of the “bottom” carbon. The $\delta = 186.6$ signal must therefore correspond to the positively charged “end” carbons of the delocalized cation:



17. (a) $(\text{CH}_3)_3\text{CCl}$, AlCl_3 ; $(\text{CH}_3)_3\text{CCl}$, AlCl_3 ;
 (b) Cl_2 , FeCl_3 ; Cl_2 , FeCl_3 ;
 (c) H_2 , Pt ; H_2 , Pt ;
 (d) HNO_3 , H_2SO_4 ; HNO_3 , H_2SO_4 ;
 (e) CH_3COCl , AlCl_3 ; CH_3COCl , AlCl_3 ;
 (f) $\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 ; $\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 ;
 (g) SO_3 , H_2SO_4 ; SO_3 , H_2SO_4 ;
 (h) Br_2 , FeBr_3 ; Br_2 , FeBr_3 ;
 (i) chlorocyclohexane, FeCl_3 . FeCl_3 .

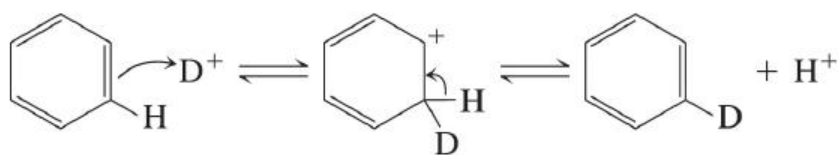
18.





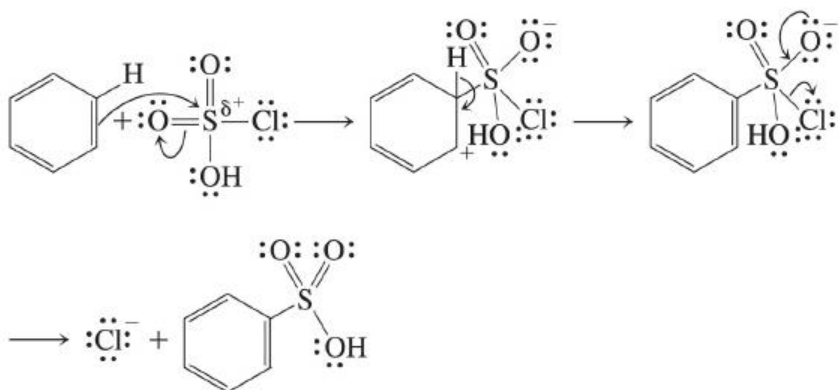
(f) is shown in the answer to [Problem 48](#).

50. The problem is twofold: What should we make C_6D_6 from, and how should we do it? The chapter does not give us any practical way to construct benzene rings from nonbenzenoid starting materials, but it does present methods to replace groups on benzene rings—electrophilic aromatic substitution. We consider what we want to accomplish: the attachment of deuterium, D , to every carbon. A plausible electrophile to consider is the deuterium ion, D^+ . In [Section 15-10](#), we see that hydrogen ions, H^+ , are sufficiently electrophilic to attack benzene rings (in that case, benzenesulfonic acid, leading to replacement of the $-\text{SO}_3\text{H}$ group by $-\text{H}$). We could therefore expect that D^+ would behave similarly, attacking benzenesulfonic acid and replacing the $-\text{SO}_3\text{H}$ group by $-\text{D}$. Indeed, it does, but is that really the best way to solve our problem? That reaction in [Section 15-10](#) was presented to show how to get rid of the $-\text{SO}_3\text{H}$ group. But do we need that group in the first place for what we're trying to accomplish here—replace H by D ? The chemistry in [Section 15-10](#) tells us that D^+ can attack a benzene ring as an electrophile. Therefore, we can assume that it can attack benzene itself:

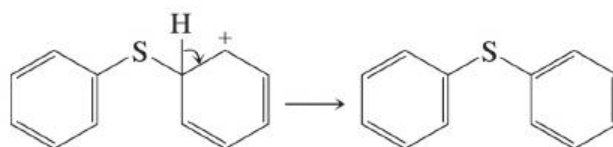
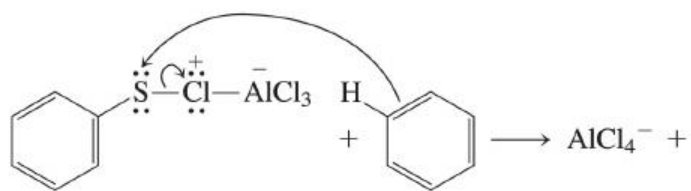
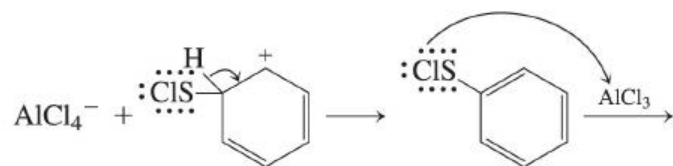
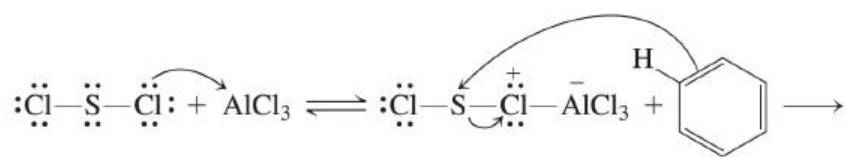


Run in the forward direction, we've replaced an HH with a D!D! . But the reaction is an equilibrium: We need to drive it from left to right. The solution is to treat benzene with a large excess of an acidic solution containing $\text{D} + \text{D}^+$ instead of $\text{H} + \text{H}^+$, dilute D_2SO_4 in D_2O , for example. The reaction goes to equilibrium, replacing most of the benzene $\text{C}-\text{HC}-\text{H}$ bonds with $\text{C}-\text{DC}-\text{D}$ bonds, according to the equation above. We repeat the process several times, each time treating the partially deuterated benzene with fresh D_2SO_4 in D_2O , until the amount of residual HH in the benzene has been reduced to an acceptably low level (typically well below 1%).

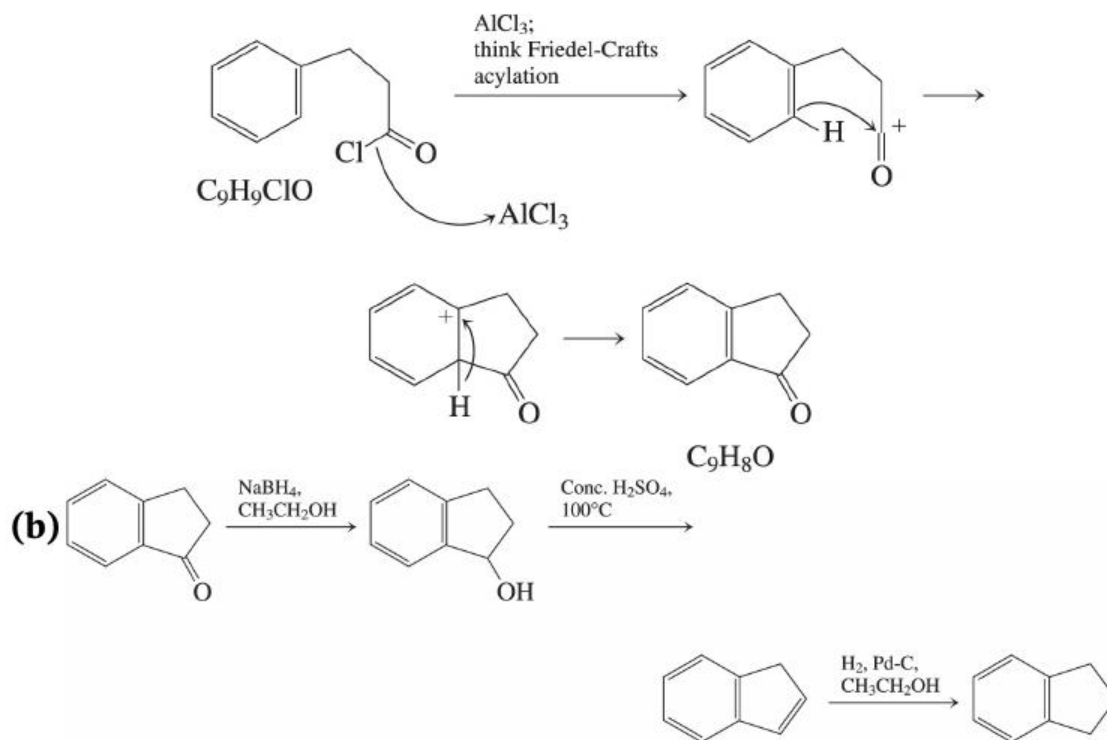
51. Identify a likely electrophilic atom and follow a reasonable mechanistic pattern.



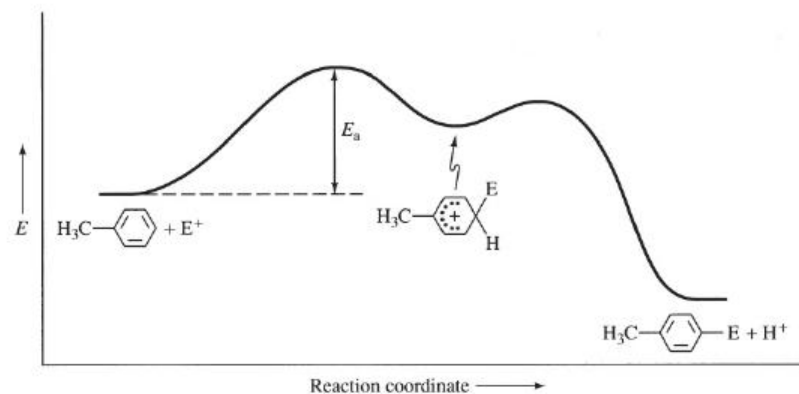
52. Consider a mechanism similar to Friedel-Crafts alkylation. Twice.



53. (a) Think mechanistically. Then see if what you come up with matches the data.



54.

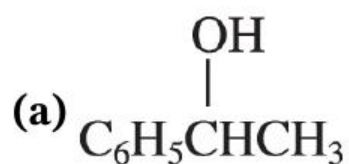


Schore, *Study Guide and Solutions Manual for Organic Chemistry*,
8e, © 2018 by W. H. Freeman and Company

The reaction of methylbenzene should proceed with a lower energy of

activation than that of benzene ($E_{\text{a}}^{\text{methylbenzene}} < E_{\text{a}}^{\text{benzene}}$), and the intermediate cation should be more stable.

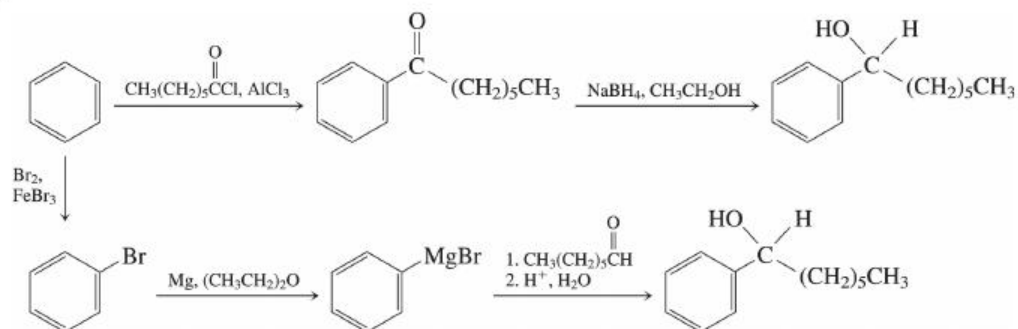
55.



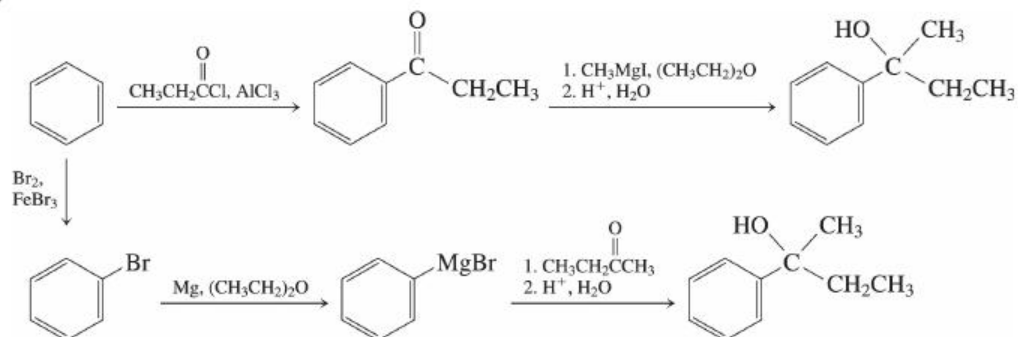
56. More than one approach can be used for both (a) and (b): one based on Friedel-Crafts acylation, which uses an alkanoyl chloride, and another using Grignard addition to an aldehyde or a ketone. If you got one but not the other, try now to give a second answer before peeking at the solutions below.

Note: Aqueous acid work-up for Friedel-Crafts acylation is standard and is omitted from this and subsequent schemes.

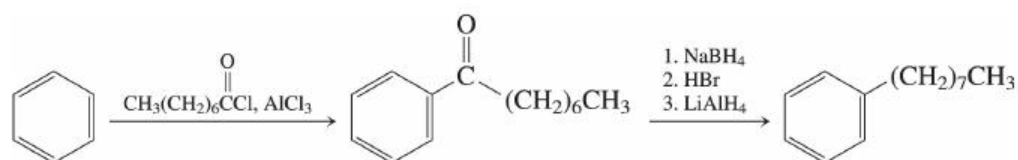
(a)



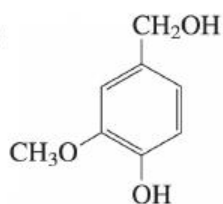
(b)



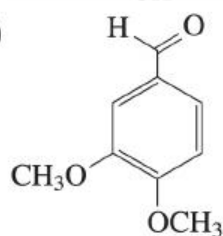
(c) Friedel-Crafts alkylation is liable to lead to rearranged products. Use acylation-reduction.



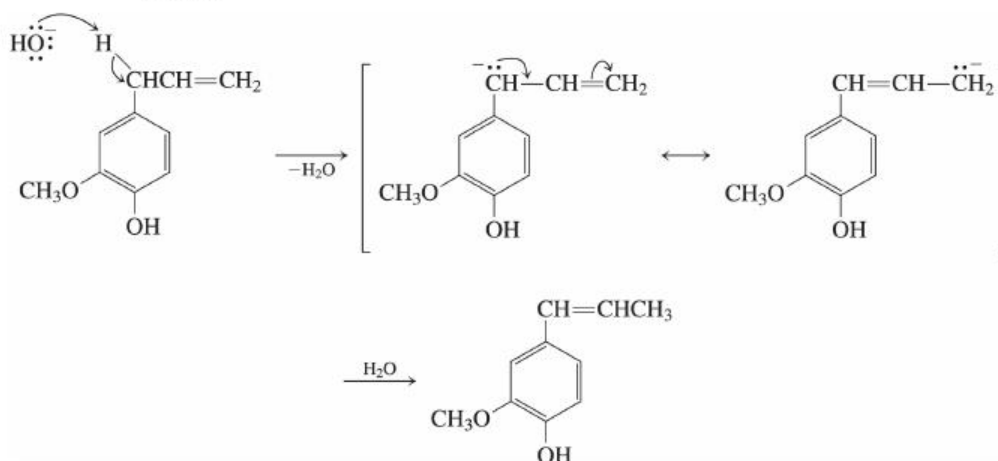
57. (a)



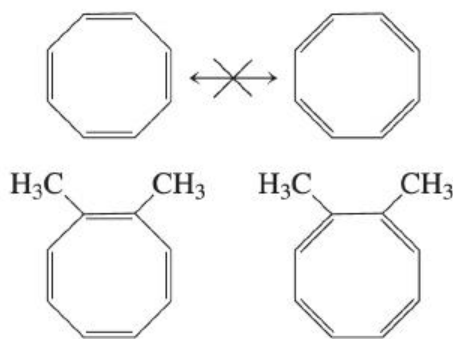
(b)



(c)

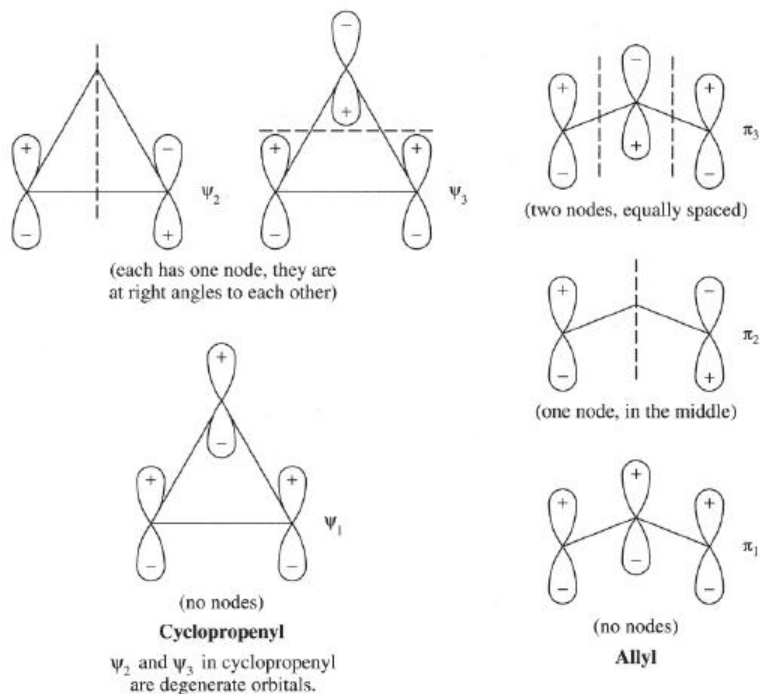


58. Cyclooctatetraene lacks resonance stabilization; its double bonds behave as if they were isolated, not conjugated. This is in fact the case. As a result of the geometry of the molecule (which is **not** planar; see Figure 15-17), the double bonds do not overlap into conjugated systems. So, resonance does **not** occur:



The two structures above actually represent **different** molecules!
 Their names are 1,2-dimethylcyclooctatetraene and 1,8-dimethylcyclooctatetraene, respectively.

59. (a) Draw the molecular orbitals by referring to the ones in Figure 15-4 and putting the **nodes** in your orbital illustrations the same way.



Schore, *Study Guide and Solutions Manual for Organic Chemistry*,
 8e, © 2018 by W. H. Freeman and Company

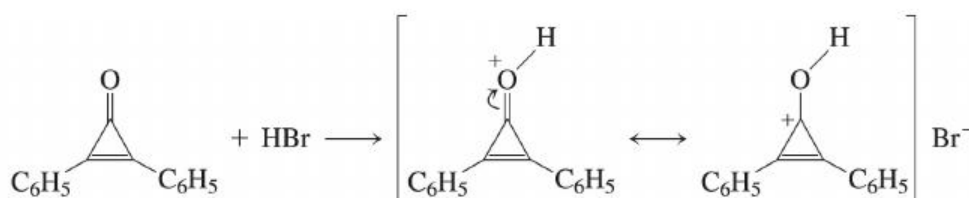
- (b) **Two** electrons are best. They would fill the cyclopropenyl ψ_1 orbital and be stabilized relative to electrons in the π_1 allyl orbital. Electrons in ψ_2 or ψ_3 of cyclopropenyl are destabilized relative to π_2 electrons in allyl. Lewis structures for the two-electron cyclopropenyl and allyl systems (both are cations):



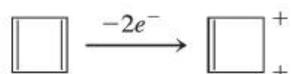
- (c) Yes! The cyclopropenyl cation has two $\pi\pi$ electrons in a complete.

unbroken “circle” (really, triangle) of p orbitals, so it fits Hückel’s rule. These electrons can be represented as being delocalized in a cyclic manner, and the system is stabilized relative to the best acyclic analog, the 2-propenyl (allyl) cation.

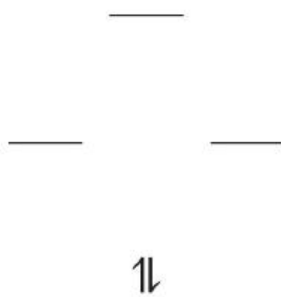
30. Although there are several locations in this molecule that are reasonable sites for protonation, one in particular, the Lewis basic carbonyl oxygen, is unusual: Upon protonation, a resonance-stabilized cation is formed, and one of the resonance forms can be recognized as a substituted cyclopropenyl cation, an aromatic species ([Problem 59](#)). In fact, with the additional phenyl substitution, this species is remarkably stable.



31. The cation is derived by removing two $\pi\pi$ electrons from the neutral diene:

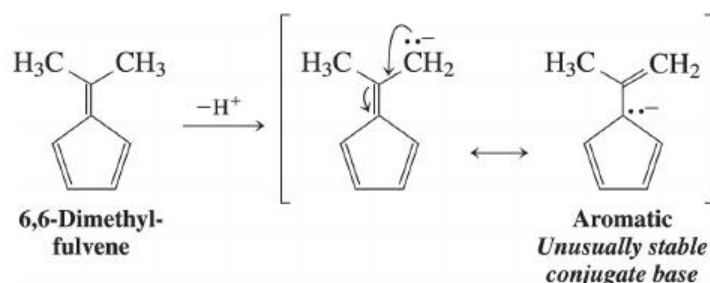


So the $\pi\pi$ system of the ring contains $2\pi 2\pi$ electrons total. That makes it aromatic according to Hückel’s rule. The MO MO levels look like this:



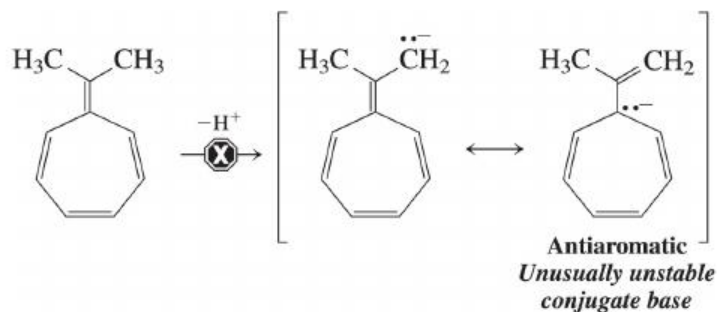
32. Recall that unusually strong acidity correlates with unusual stability of the conjugate base. A common way to increase the acidity of a hydrogen atom in a molecule containing only carbon and hydrogen is for the conjugate base to be stabilized by aromaticity. Use this principle and find the structure in which removal of a proton from one of the carbon atoms gives rise to an aromatic conjugate base. (**Hint:** Look for aromatic five-carbon rings in [Section 15-7](#).)

- (a) The cyclopentadienyl anion is aromatic. Only one of the four fulvenes pictured gives an aromatic cyclopentadienyl anion upon loss of a proton, by means of resonance delocalization of the lone pair left behind:



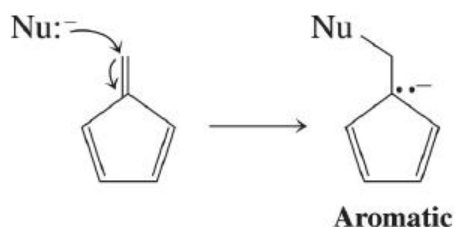
Therefore, the compound shown, 6,6-dimethylfulvene, is the one that is unusually acidic.

- (b) The corresponding seven-membered ring compound shown below would give an 8π -electron, 8π -electron, *antiaromatic* anion upon deprotonation of one of its methyl groups. This would be a very unstable conjugate base and, therefore, the parent hydrocarbon is a very *weak* acid.



33. Again, as in Problem 62, look for aromaticity to play a role. Addition

of a nucleophile to the non-ring carbon gives an aromatic result:

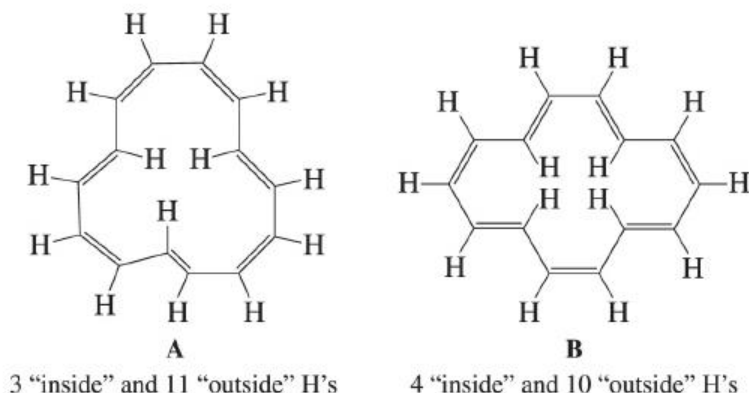


In contrast, addition of a nucleophile to any of the five ring carbons makes that carbon tetrahedral, breaking up the continuous circle of p orbitals necessary for aromaticity. Therefore such additions are normally not observed.

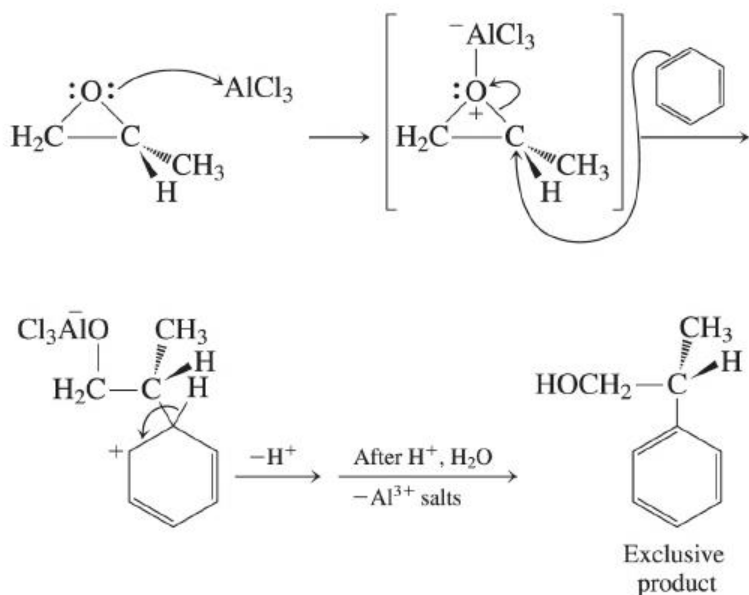
34. Figure 15-9 illustrates the local magnetic field generated by the ring current of an aromatic molecule. As described in [Section 15-4](#), h_{local} reinforces H_0 outside the ring, a reinforcement leading to deshielding of protons on the external periphery. In contrast, h_{local} points in the **opposite** direction at the **center** of the ring and therefore reduces the net field in this region. In aromatic rings large enough to possess hydrogens inside the ring of circulating π electrons, such hydrogens should be strongly shielded, requiring increased values of the external field H_0 to attain resonance. Their NMR signals should be shifted to high field (to the right).

- (a) The structure of [18]annulene ([Section 15-6](#)) shows 12 hydrogens outside the ring (blue), which will be deshielded as are those of benzene. The inner six (green) are strongly shielded, a situation resulting in the signal nearly 3 ppm upfield from that of $(\text{CH}_3)_4\text{Si}$.
- (b) Yes. The eight hydrogens on the π periphery show benzenelike deshielding, whereas the two hydrogens on the bridge over the ring are shielded, similar to those “inside” the [18]annulene ring.

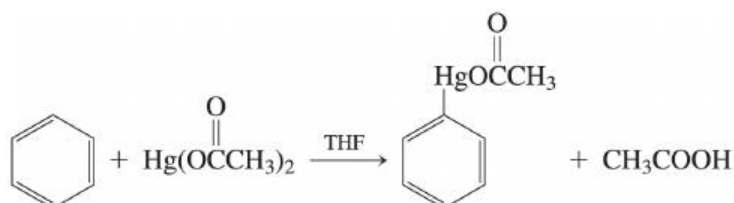
35. Structure A contains four cis and three trans double bonds; structure B contains three cis and four trans double bonds. The NMR indicates the presence of four “inside” and ten “outside” hydrogens, a pattern consistent with structure B only.



36. Look at AlCl_3 as a Lewis acid and benzene as a nucleophile. Recall ([Section 9-9](#)) that acid-catalyzed ring openings of oxacyclopropanes give $\text{S}_{\text{N}}1$ regiochemistry (most stable carbocation) but $\text{S}_{\text{N}}2$ stereochemistry (backside nucleophilic attack). So,

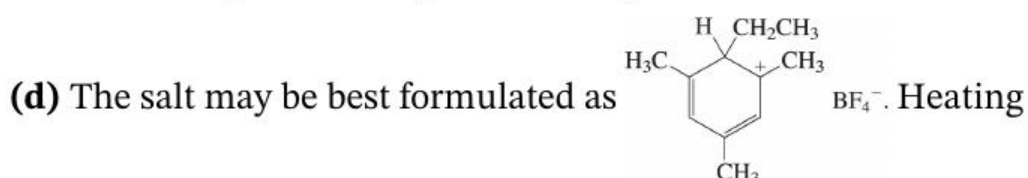


37. How about a simple electrophilic substitution? Hg_2^{2+} in $\text{Hg}(\text{OCCH}_3)_2$ is electrophilic, so try



38. (a) In analogy to the reaction of alkyl halides, such as RCl , with AlCl_3 , a reaction between HCl and AlCl_3 might be expected to occur, too. Recall ([Section 15-11](#)) that either of two types of reactions takes place with RCl : A Lewis acid-Lewis base complex forms with primary RCl , but secondary and tertiary RCl ionize to give carbocations. We know that HCl is easily ionized, so it makes sense to suggest that ionization occurs upon reaction with AlCl_3 : $\text{HCl} + \text{AlCl}_3 \rightarrow \text{H}^+ + \text{AlCl}_4^-$. The solution of ions conducts electricity.
- (b) The largest chemical shifts belong to C_1 , C_3 , and C_5 , which make sense because these happen to be the ones that bear positive charge in the three resonance forms of this cation, and should be most deshielded. C_2 and C_4 have pretty normal chemical shifts for benzene carbons. C_6 has a chemical shift in the range for sp^3 -hybridized carbon atoms, as expected from the structure.
- (c) Two effects are indicated by the table: First, each additional methyl group added to the ring increases the rate of electrophilic aromatic substitution by a significant amount. This observation is easy to rationalize, because methyl groups are electron releasing and should inductively stabilize the carbocationic intermediate in the reaction and in turn lead to a lower energy transition state.

the reaction and, in turn, lead to a lower-energy transition state leading to the cation. Second, in the three instances for which rate data are given for two *different* isomers with the *same number* of methyl groups, the rates for the two isomers are not the same. The effect is small, about a factor of two, for di- and trimethylbenzenes. In the case of the dimethylbenzenes, the reason is steric: In each molecule the carbocations are similarly stabilized. However, all positions in 1,4-dimethylbenzene are adjacent to a methyl group, while substitution at positions 4 or 5 in 1,2-dimethylbenzene avoids the steric crowding of an adjacent CH_3 . For the trisubstituted benzenes, the 1,2,3-isomer is faster because substitution at either of two positions, C4 or C6, gives a carbocation in which two of the three resonance forms benefit by placing the positive charge on a tertiary ring carbon. In contrast, in the 1,2,4-isomer, substitution at C3 is sterically difficult because it lies between methyl groups on either side. Finally, the spectacular rate difference for the two tetrasubstituted benzenes requires explanation. The difference here is that substitution in the 1,2,3,5-isomer proceeds through an intermediate in which all three positions that bear positive charge are tertiary. In the 1,2,3,4-isomer, no intermediate has more than two tertiary sites with positive charge.



causes elimination of HBF_4 and generation of the neutral, tetrasubstituted benzene. Why is the salt stable in the absence of heat? Make a model. The sp^3 carbon atom holds the bulky ethyl group out of the plane of the methyl groups on either side. Upon loss of the proton, the ethyl group moves into the same plane as these methyl groups and experiences steric hindrance as a result. The benzene aromaticity is enough to thermodynamically overcome the unfavorable steric effect, but the latter clearly shows up in a higher than usual activation barrier for proton loss and

up in a higher than usual activation barrier for proton loss and aromatization.

69. (a)

70. (c)

71. (d)

72. (b)

73. (d)

16

Electrophilic Attack on Derivatives of Benzene: Substituents Control Regioselectivity

The preceding chapter introduced you to the properties and chemistry of benzene itself. In this chapter the chemistry is expanded to include several new reactions not only of benzene but also of derivatives of benzene containing various substituents. The text focuses on the effects of substituents on the chemistry of the benzene ring. The first and rate-determining step of electrophilic substitution is the formation of a delocalized cation. Different types of substituents can affect the stability of this cation and make its formation easier or harder. Substituents on the ring also direct the attack of an electrophile to specific positions. This study guide chapter contains information that may help you avoid a lot of memorization. You will see that the effect of a substituent on benzene chemistry can be predicted from a relatively straightforward examination of its structure and its consequential electronic characteristics.

Outline of the Chapter

[16-1 Activation or Deactivation by Substituents on the Benzene Ring](#)

[16-2 Directing Inductive Effects of Alkyl Groups](#)

16-3 Effects of Substituents in Conjugation with the Benzene Ring

16-4 Electrophilic Attack on Disubstituted Benzenes

Effects of substituents on the reactivity and regioselectivity of further electrophilic substitution on a benzene ring.

16-5 Keys to Success: Synthetic Strategies

General principles and useful new reactions.

16-6 Reactivity of Polycyclic Benzenoid Aromatics

What happens when two or more benzene rings are fused together.

16-7 Polycyclic Aromatic Hydrocarbons and Cancer

Keys to the Chapter

16-1 through 16-3. Directing Effects: Activation and Deactivation

Don't forget the basics: **Electrophiles** seek **electrons**! So, reactivity of a benzene ring toward electrophiles will be enhanced by substituents that donate electron density to the ring, and it will be reduced by substituents that withdraw electron density from the ring. "Reactivity" in this context is the **rate** at which substitution occurs, which is determined by the rate-determining step—electrophilic attack to form a carbocation. Electron-donating groups (*activating groups*) will stabilize the cation, lowering the transition state energy for its formation and speeding up the reaction, and electron-withdrawing groups (*deactivating groups*) will do just the opposite. Furthermore, these effects will differ at different positions around the ring relative to the location of the substituents already present, giving rise to *directing effects*.

Both inductive and resonance effects are involved. The favored reaction proceeds through the most stabilized (or least destabilized) intermediate carbocation. Study **carefully** the resonance forms pictured for the possible cations derived from electrophilic attack on methylbenzene (toluene) and (trifluoromethyl)benzene ([Section 16-2](#)), and on aniline, benzoic acid, and a halobenzene ([Section 16-3](#)). Notice the **types** of groups that fall into the different categories in Table 16-1. In particular, notice the following two general trends:

- .. **All** substituents with **lone pairs** on the atom attached to the benzene ring are **ortho, para** directors.
- !. **All** substituents that have a **positively polarized (δ^+) atom** lacking a lone pair attached to the benzene ring are **meta** directors.

These generalizations always hold and can be used as aids to help you remember whether a group belongs in one category or the other.

16-5. Keys to Success: Synthetic Strategies

This section shows how to plan syntheses of multiply substituted benzenes. It also illustrates the interconversion of NH_2 and NO_2 , reduction of $\text{C}=\text{O}$ to CH_2 , use of SO_3H as a para-blocking group, and several other valuable synthetic “tricks.”

16-6. Reactivity of Polycyclic Benzenoid Aromatics

Fused polycyclics such as naphthalene, anthracene, and phenanthrene exhibit both similarities and differences with respect to benzene. For example, electrophilic aromatic substitution occurs but is generally easier with the polycyclics and proceeds with milder reagents. The activation energies for attack are lower because the intermediate cations are more highly delocalized (more resonance forms!) and, therefore, lower in energy. If the rings are not identically substituted, reaction occurs at the most activated ring. Substituents have the same directing and activating/deactivating effects you saw with benzene.

Fused polycyclics also have an increased tendency to undergo **addition** reaction, which are rare for benzene itself. As a general rule of thumb, additions occur in such a way as to preserve intact benzene rings as much as possible. So, additions to both anthracene and phenanthrene occur at the 9,10 positions, leaving two simple benzene rings intact.